

AI Deep Research Agent (Apify + OpenAI o3)

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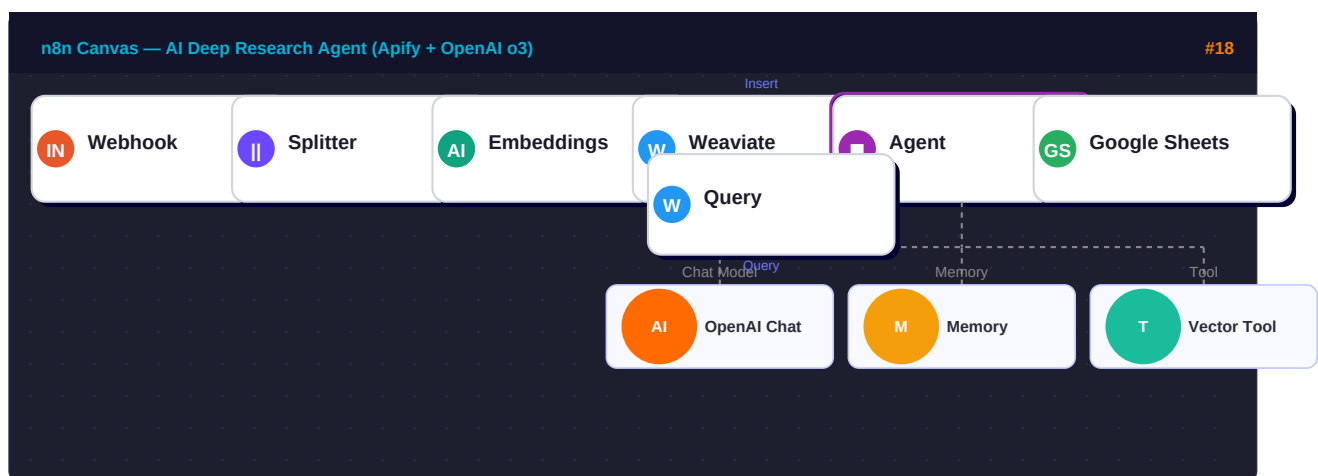
Research / Agentic Deep Research

Self-hosted Perplexity Pro — multi-source research
with full citations at \$0.10/report

BUSINESS PROBLEM THIS SOLVES

This self-hosted research agent crawls 20-50 web sources using Apify, reasons with OpenAI o3 extended thinking, and produces fully cited research reports — on your own infrastructure at near-zero variable cost per report.

n8n CANVAS FLOW — node layout from imported workflow JSON



STEP-BY-STEP — what each node actually does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — research question	POST: {question, depth: deep/quick, focus_domain, output_format}
2	⌂ Splitter — background context docs	Pre-loaded domain docs split for RAG background knowledge
3	■ OpenAI Embeddings — internal KB	Background documents indexed. Agent supplements web results
4	■ Weaviate — internal knowledge store	Internal docs and previous research reports stored
5	■ Weaviate Query — existing knowledge	'What do we already know about 5G spectrum in Oman?' — avoids redundant work
6	■ Vector Tool — background retriever	Agent tool for accessing internal KB during research loop
7	■ Window Memory — research progress	Tracks: sources visited, findings so far, remaining gaps
8	■ OpenAI o3 — research orchestration	Extended thinking: decomposes question → search plan → evaluates credibility → fills gaps

#	WHAT HAPPENS	PRACTICAL DETAIL
9	■ Agent — web crawl + synthesise	Iterates: Apify crawl → extract facts → cite sources → identify gaps → re-search → build report
10	■ Sheets — source audit log	All URLs visited, credibility_score, key_extracts, used_in_report

SAMPLE OUTPUT — what the workflow actually produces

"""5G Mid-Band Spectrum in GCC — Research Report Oman: TRA auctioned 3.5GHz to Ooredoo and Omantel in Q4 2023 [1]. UAE: TDRA awarded 3.5GHz to du and e& in 2021 [2]. Sources: [1] TRA Oman Dec 2023 [2] TDRA UAE 2021 Annual Report"

CREDENTIALS REQUIRED — practical setup guide

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key — o3 tier	platform.openai.com — o3 model requires API Tier 4+	OpenAI API
Apify API Key	console.apify.com → Settings → Integrations → API token	Apify API
Weaviate Cloud URL + Key	console.weaviate.cloud → Create cluster	Weaviate API

PORTFOLIO RELEVANCE — why this matters for GCC AI roles

Flagship agentic showcase — demonstrates the Deep Research pattern that G42, SDAIA, and N2S.Global AI Lead interviews specifically probe.

JSON: deep_research_agent.json	GitHub: trm-deep-research-agent	Ref: apify.com/...
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